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Imaging dosimetry

An alternative approach to account for patient organ doses from imaging guidance procedures

Alan P. Nelson [a], George X. Ding [a,b,*]

[a] Department of Radiation Oncology, Vanderbilt University School of Medicine; and [b] Department of Physics and Astronomy, Vanderbilt University, Nashville, USA

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ABSTRACT

Purpose: To investigate the feasibility of an alternative method of accounting for additional organ doses resulting from image guidance procedures during patient treatment planning through tabulated values based on scan protocol and scan site.

Methods and materials: Patient-specific imaging dose to 30 patients resulting from Varian OBI kV–CBCT scans using the Standard Head (17 patients), Low-dose Thorax (8 patients), and Pelvic (5 patients) scan protocols were retrospectively calculated using Monte Carlo methods. Dose dependence on scan location and patient geometry was explored. Patient organ doses were analyzed by using dose–volume histograms and expressed by the mean, minimum dose delivered to 50% of the organ volume, D50. The reported doses are dose-to-medium instead of dose-to-water.

Results: The organ doses from all patient-specific calculations show predictable and limited ranges across patients. For brain isocenters using Standard Head Scans: Bone: 0.7–1.1 cGy, Brain: 0.2–0.3 cGy, Brainstem: 0.2–0.3 cGy, Skin: 0.3–0.4 cGy, Eye: 0.03–0.3 cGy. For head and neck patients using the Standard Head Scan: Bone: 0.3–0.6 cGy, Parotids: 0.3–0.4 cGy, Spinal Cord: 0.15–0.25 cGy, Thyroid: 0.1–0.25 cGy, Skin: 0.2–0.3 cGy, Trachea–Esophagus: 0.1–0.2 cGy. For chest using Thorax Scans: Bone: 1.1–1.8 cGy, Soft tissue organs (Bowel, Lung, Heart, Kidney, Esophagus, and Spinal Cord): 0.3–0.6 cGy. For abdominal site using Pelvic Scans: Bone: 3.2–4.2 cGy. Soft tissue organs (Bladder, Bowel, Rectum, Prostate, and Skin) D50s fell between 1.2 and 2.2 cGy. Femoral Heads: 2.5–3.4 cGy.

Conclusions: It is adequate to estimate and account for organ dose by using tabulated values based on scan procedure and site because organ doses from imaging procedures are only modestly dependent upon scan location and body size. Considering the dose variation and magnitude of dose from each scan protocol in comparison to therapeutic doses, this approach provides a simple alternative to account for additional imaging guidance doses during patient treatment planning. Clinicians can use these tabulated values to make informed decisions in selecting the appropriate imaging procedures and imaging frequency during radiotherapy treatment.

Image-guided radiation therapy (IGRT) is widely used on a daily basis with increased use of kV–CBCT for patient alignment. IGRT undoubtedly has many important benefits such as enabling better tumor localization and reduced margins, but so far the question of how to take into account and report the extra imaging dose in clinical practice has not been clearly addressed.

Among the commonly used forms of image procedures in IGRT are kilo-voltage radiographs and cone-beam computed tomography (kV–CBCT). kV imaging provides superior image contrast compared with MV portal imaging. While the dose from a single imaging procedure is much lower than a fraction of therapeutic dose, the summed dose of repeated imaging procedures may not be negligible and should be accounted for [1]. The main approaches used to determine this imaging dose are experimental measurements [2–11], Monte Carlo methods [12–22], and simplified kV imaging dose calculation modules integrated into a treatment planning system [23].

With the increased interest in managing the additional imaging dose to radiotherapy patients, a practical method of accounting for the additional dose to patients resulting from these imaging procedures has yet to be developed. Although measurements can be done, performing measurements for each patient would be very difficult in a clinical setting. Although accurate patient-specific kV dose calculations are best achieved using Monte Carlo methods, the long calculation times and expertise required by Monte Carlo

* Corresponding author at: Department of Radiation Oncology, Vanderbilt University Medical Center, 1301 Medical Center Drive, PRB, B1031, Nashville, TN 37232-5671, USA.

E-mail address: george.ding@vanderbilt.edu (G.X. Ding).

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simulations make it impractical in a clinical setting to calculate imaging doses for every patient. Most current treatment planning systems are incapable of calculating and accounting for kV imaging dose for specific patients. This limitation is due to the fact that scatter properties and increased photoelectric interaction in the diagnostic X-ray energy range make it far more complex to accurately calculate doses when compared with a therapeutic MV beam. Those treatment planning systems that do include a method of kV dose calculation are limited in accuracy, especially in and around bony anatomy [23,24]. Moreover, these calculations still take a fairly long time to complete, and therefore may not be appropriate in a clinical setting.

Given the relatively small magnitude of dose from each image guidance procedure, it may be adequate to use simpler approaches that provide reasonably accurate estimates of the imaging dose. The purpose of this study is to use accurate patient-specific imaging dose calculations over a wide range of patients to provide organ dose estimates based on scan protocol and scan site. If inter-patient variation and geometry dependence is found to be reasonably small, these dose range estimates would provide simple look-up values that may be accurate enough to account for the additional dose from the imaging procedures. The simplicity of this approach would enable clinicians to easily integrate it into their clinical workflow.

Methods and materials

The image procedures using Varian OBI (1.4/1.5) devices for kV–CBCT acquisitions were investigated. In all cases the Varian default clinical scan protocols were used. The specific scan parameters are listed in a previous study [15]. It is worthy to note that OBI clinical default protocols are generally used in daily clinical practice. There are 30 patient data sets that were selected for the study including 9 brain patients, 8 head and neck patients, 8 lung and chest patients and 5 pelvic patients. These data sets are selected in order to cover a representative range of patients encountered in the clinic with regard to the patient size and treatment target locations. We also studied the impact of variations in scanned isocenter location for the same data set.

Monte Carlo calculations

Patient CTs were used as inputs to Monte Carlo simulations to retrospectively calculate 3D imaging dose distributions from the Varian OBI v1.4 kV–CBCT scans. Although there were uncertainties due to the intrafraction and interfraction organ motion between planning CT and kV–CBCT, positional changes in the order of a centimeter should make only a small difference in the dose received by an organ because kV–CBCT involves a broad beam. The Monte Carlo dose calculations consist of the Monte Carlo simulation of each individual kV source, calibration of each kV source, and calculations of the dose due to each kV imaging procedure. BEAMnrc [25] was used to simulate kV X-ray sources. The X-ray tube model included the anode materials and target angle, the filament location, the X-ray window materials, and the collimation system. The simulated beam of each scan protocol was stored in phase space files and used in the DOSXYZnrc [26] simulation. Each simulated beam was validated against measurements [27]. The energy thresholds for secondary particle creation (AE, AP) and energy cutoff (ECUT, PCUT) for particle transport [28] were set to AE = ECUT = 0.516 MeV (total electron energy) for electrons and AP = PCUT = 0.001 MeV for photons.

The calibration of each kV source was performed by dose measurements. The dose to water, $D_w^{chamber}$, determined by using an ionization chamber is based on the following equation [29]:

$$D_w^{chamber}\ (kV) = MN_k P_{Q,cham} \left(\frac{\overline{\mu}_{en}}{\rho}\right)_{air}^{w}$$

where D_w is the dose to water in the phantom, M is the leakage-corrected chamber reading in Coulombs (C) and corrected for other influencing quantities including temperature and pressure, N_k is the air kerma calibration factor for the given beam quality, Q, in Gy/C, $P_{Q,cham}$ is the overall chamber correction factor that accounts for the change in the chamber response due to the displacement of the phantom by the ionization chamber and the presence of the chamber stem, the change in the energy, etc., and $\left(\frac{\overline{\mu}_{en}}{\rho}\right)_{air}^{w}$ the water-to-air ratio of the mean mass energy-absorption coefficients. The air kerma calibration factors for X-ray beam quality specified by half-value layer (HVL) and kVp were obtained from an Accredited Dosimetry Calibration Laboratory (ADCL) [29,30]. The phantom used to determine the CBCT output is water equivalent, Plastic Water® low energy range (PW-LR) (CIRS, Norfolk, VA) slabs with a size of $30 \times 6 \times 30\ \text{cm}^3$ and the ion chamber, Exradin Model A14, was inserted into the center of the phantom. The potential errors in performing the beam output calibration for Monte Carlo calculations in the PW-LR phantom were shown to be within 2% [27]. The Monte Carlo calculated dose has been validated using the method reported in previous studies [13,22].

Patient specific dose calculations were done using the code DOSXYZnrc [26] by rotating the X-ray source around the patient based on the specific scan procedure. The calculation voxel size was $2.5 \times 2.5 \times 2.5\ \text{mm}^3$.

In general, the patient's treatment isocenter was used in all cases as the kV–CBCT isocenter. To specifically analyze isocenter dependence of imaging dose, a brain patient and a prostate patient was simulated with 3 separate isocenters. These isocenters were 5 cm apart and 8 cm apart in the anterior–posterior direction for the brain and prostate patient, respectively.

It is worth to note that the Monte Carlo calculated absorbed doses are dose-to-medium and not dose-to-water in the respective structure. Over the last decades, the doses calculated in model based commercial treatment planning systems are dose-to-water. Although differences are small for soft tissues, the dose to bone is two to four times greater than dose to soft tissue for kV X-rays [14,18].

Organ dose analysis

The patients analyzed for this study were initially contoured for treatment planning purposes using Varian Eclipse®. Skin contours were split into four quadrants to analyze entrance dose directional dependence. Organ contours were cut off to only include that portion that was "in-field" in the *z*-direction ±9.0 cm for the Head Scan and ±10.0 cm for the Pelvic and Thorax Scan. An in-house Python program was developed and used to analyze the calculated dose distribution. This program imports the Monte Carlo calculated 3D dose matrix and the DICOM structure set and performs DVH analysis by associating dose voxels with their corresponding structures.

The main quantities derived and reported from the DVHs in this study are the mean dose, the minimum dose received by 50% of the in-field organ volume (D50) and the minimum dose received by 10% of the in-field organ volume (D10). The difference between D50 and mean dose was also analyzed for this study. D10 was reported in order to provide an estimation of maximum organ doses resulting from the imaging study. These quantities are selected because they can be used to estimate organ dose effectively.

We chose to contour organs in the *z*-direction within ±9 cm and ±10 cm of the isocenter for head and pelvic/Thorax Scans, respectively. These cut-offs have been chosen in order to accurately report the organ dose because the parts of the organ that are located outside (beyond ±9 cm and ±10 cm of the scanned isocenter) are not exposed to primary X-ray beams and doses resulting from scattered radiations are much lower. These cut-off locations are set relative to each patient's specific isocenter.

To validate the accuracy of the in-house Python program, a 3D dose matrix was generated with known dose values assigned to specific locations. Eclipse was used to contour structures with a known geometry and location such that the resulting DVH could be predicted. When the known dose matrix and structure set was analyzed using the in-house program, the resulting DVH was as predicted and validated.

Results

Standard Head Scan protocol

Fig. 1 presents inter-patient variation in organ dose for patients using the Standard Head Scan protocol for brain tumor site and head and neck tumor site scans.

The results of 9 brain patient scans are shown in upper part of Fig. 1. The ranges for the organ D50s and D10s across all patients simulated with this protocol are shown in Table 1. These data show that all soft-tissue organs fell between approximately 0.1–0.4 cGy for both the D50 and D10. The one exception to this was the D10 for skin, which was 0.6 cGy in the most extreme case.

The positional dependence of dose to an organ is well observed in Fig. 1 (upper graph). The most extreme organ in terms of positional variation and dose variation in this case is the skin. While the mean or D50 for the skin as a whole does not vary greatly across patients for the Standard Head Scan (200° rotation) (0.2–0.4 cGy), different skin volumes receive a different dose depending upon their position.

The dependence of organ dose on isocenter location was explored by simulating the same patient with the isocenter in three different positions: the original isocenter and 5 cm anterior and posterior to that isocenter. The variation in D50 organ doses with these shifts was generally less than 0.1 cGy, the most variation being seen in the bone (0.5 cGy variation in D10) and eye dose (0.15 cGy variation due to the 200° rotation scan geometry). Highly anterior scan isocenters result in more eye dose, which explains the range in D50 eye doses that was observed: 0.03–0.31 cGy. Aside from the eye dose, however, the 10 cm isocenter range that was analyzed in this study did not show much variation for the other organs.

Size dependence for the Standard Head Scan was explored by including two pediatric patients in the study. The pediatric patients comprised patient #8 and #9 in Fig. 1 (upper graph). It can be seen that there is generally more dose to the internal organs relative to the adult patient, while the absolute magnitude of the dose is still quite small. The fact that the pediatric patients have the highest eye dose is mainly due to the highly anterior isocenter and the 200° scan geometry rather than the smaller head size. From the D10 bone dose, it appears that the maximum dose to the pediatric patients is actually on the lower end of the range; this is most likely due to the fact that maximum bone dose in the skull is mostly due to SSD effects, and the smaller size of the pediatric head results in most of the skull having a larger SSD over the whole scan than a patient with a larger head.

The results of 8 head and neck patient scans are shown in the lower part of Fig. 1 where the mean, D50 and D10 values for the head and neck patient isocenters are presented. These data demonstrate the combined effect of isocenter location and patient geometry dependence. The patient dose data shown is sorted by cervical vertebrae level of the isocenter which would generally result in higher soft tissue organ dose for more superior isocenters due to the smaller size of the neck relative to the shoulder region. However, different patient sizes resulted in some of the more superior isocenters receiving somewhat less organ dose. Internal organ doses are very similar to those found in the brain isocenters, while

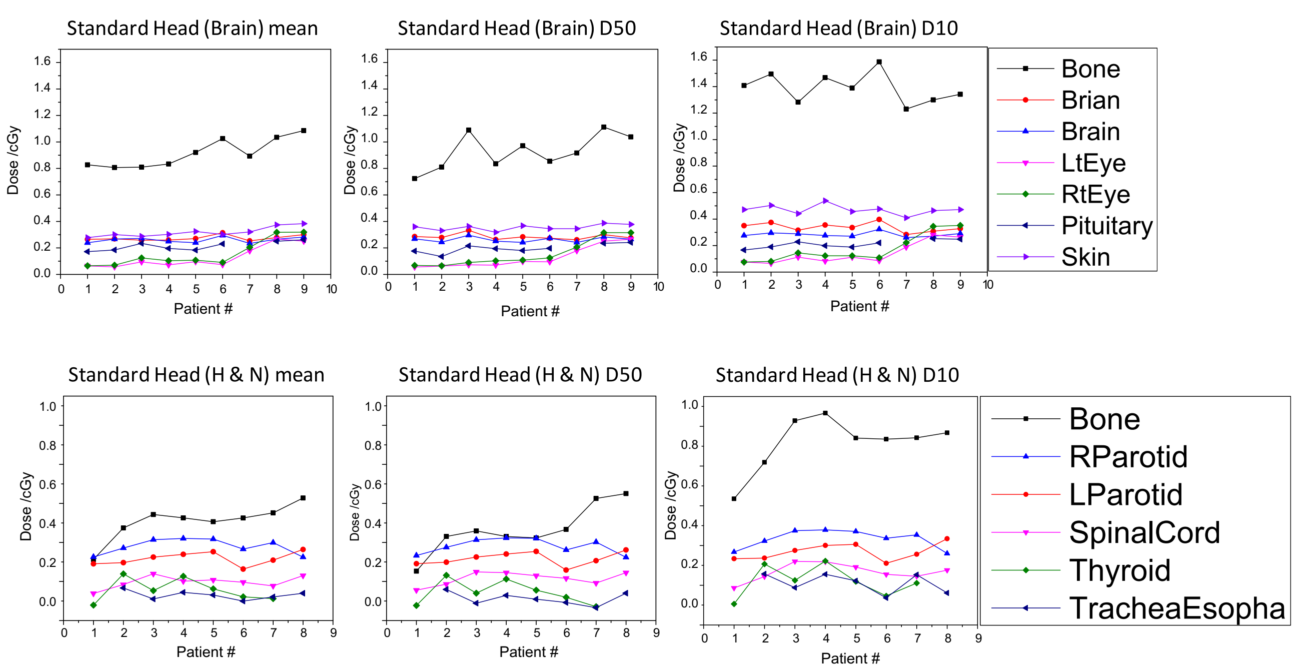

Fig. 1. Upper graph shows organ doses from the Standard Head Scan protocol for brain isocenters: mean, D50, and D10 across 9 brain patients. Patients were ordered by eye dose, with patient #8 and #9 being pediatric patients. Almost all soft tissue D50 organ doses fall between 0.1 and 0.4 cGy, with individual organ D50 doses varying less than 0.1 cGy across patients except those affected by the 200° scan geometry like the eye. Lower graph shows organ doses from the same Standard Head Scan protocol for various head and neck patient isocenters. Patients were ordered from low isocenter (level of the C5–C6 vertebrae) to high isocenter (level of C3). Lower bone dose is observed when compared to brain isocenters due to more bones being deeper. Soft tissue doses almost all fall between 0.1 and 0.4 cGy, with tighter ranges for individual organs. Note that the dose to bone is two to four times greater than dose to soft tissue for kV X-rays. The lines that connect the data points are to help separate different organs.

Table 1
Standard Head Scan D50 and D10 organ dose ranges for brain isocenter and head and neck isocenter patients. Note that the dose to bone is two to four times greater than dose to soft tissue for kV X-rays.

Standard Head, brain isocenters			Standard Head, head and neck isocenters		
Organ	D50 range (cGy)	D10 range (cGy)	Organ	D50 range (cGy)	D10 range (cGy)
Brain	0.26–0.33	0.28–0.40	Brain	0.15–0.22	0.16–0.23
Brainstem	0.24–0.30	0.26–0.32	Larynx	0.21–0.29	0.25–0.33
Chiasm	0.16–0.26	0.17–0.26	Oral Cavity	0.13–0.26	0.20–0.31
Eyes	0.03–0.31	0.04–0.35	Parotids	0.26–0.42	0.31–0.48
Optic Nerves	0.08–0.25	0.09–0.27	Spinal Cord	0.16–0.25	0.19–0.32
Pituitary	0.14–0.24	0.17–0.25	Thyroid	0.07–0.23	0.11–0.32
Spinal Cord	0.26–0.33	0.29–0.34	Esophagus	0.07–0.16	0.14–0.26
Skin	0.32–0.39	0.41–0.54	Skin	0.18–0.27	0.34–0.44
Bones	0.72–1.11	1.23–1.60	Bones	0.25–0.65	0.64–1.07

the main difference appears to be lower bone dose. This is likely due to the fact that most bones in the head and neck region are deeper in the body relative to the skull and therefore generally receive a lower dose.

Low-dose Thorax Scan protocol for chest site

Fig. 2 (upper graph) shows organ mean, D50s and D10s for the Low-dose Thorax Scan protocol (370° rotation) across 8 lung and chest patient scans and dose ranges are given in Table 2. It is seen that virtually all internal organs across all patients fell between about 0.3–0.6 cGy for D50 and 0.3–0.7 cGy for D10.

The chest scan patients shown were ordered by D50 bone dose. This ordering turned out to inversely match the average equivalent distances across patients very well. The two exceptions are patient #2, who has arms down in the scan (which caused a lower bone dose due to the relatively deep (compared to rib) humerus being included and patient #7, who had a very low isocenter that resulted in a large equivalent distance (very little lung tissue in the field) but with all the bones concentrated on the perimeter of the body (ribs/spine) as opposed to some of the more superior isocenters where the ribs are deeper in the body.

Table 2
Low-dose Thorax Scan D50 and D10 organ dose ranges for lung and spine patients.

Low-dose Thorax		
Organ	D50 range (cGy)	D10 range (cGy)
Aorta	0.42–0.58	0.44–0.63
Lungs	0.30–0.63	0.43–0.72
Small Bowel	0.33–0.54	0.39–0.61
Esophagus	0.29–0.60	0.35–0.74
Kidney	0.43–0.54	0.49–0.59
Heart	0.31–0.55	0.41–0.63
Liver	0.31–0.51	0.38–0.61
Spinal Cord	0.32–0.54	0.35–0.78
Spleen	0.32–0.52	0.36–0.60
Stomach	0.28–0.57	0.31–0.62
Trachea	0.36–0.71	0.47–1.04
Skin	0.46–0.57	0.64–0.89
Bones	1.06–1.74	1.47–2.25

Pelvic Scan protocol for abdomen site

The results of 5 pelvic patient scans are presented in the lower graph of Fig. 2 showing organ mean, D50s and D10s with organ dose ranges given in Table 3a. It is seen that virtually all internal

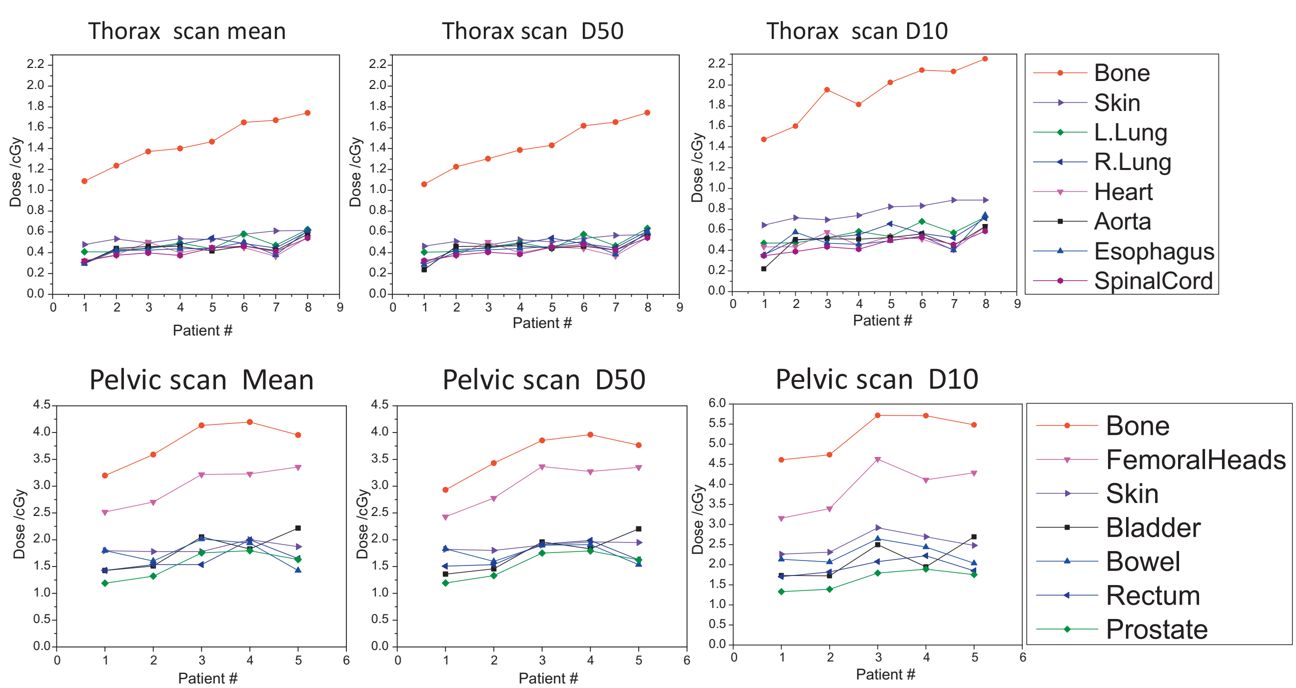

Fig. 2. Low-dose Thorax Scans across patients ordered from smallest to largest bone dose, which correlates very well with the inverse of the average equivalent thickness of the patients. Practically all soft tissue organ doses fall between 0.3 and 0.6 cGy. Very similar trends are seen for Pelvic Scan in order from largest to smallest patient size.

Table 3
(a) D50 and D10 organ dose range data for Pelvic Scan with a prostate isocenter. (b) Changes in D50 (cGy) to these organs after shifting the isocenter 8 cm posteriorly and 8 cm anteriorly.

Pelvic Scan, prostate isocenter			Pelvic Scan ant./post. shifts D50 change (cGy)		
Organ	D50 range (cGy)	D10 range (cGy)	Organ	8 cm post.	8 cm ant.
(a)			*(b)*		
Bladder	1.36–2.20	1.72–2.69	Bladder	−0.66	0.01
Bowel	1.54–1.91	2.04–2.65	Bowel	−0.64	0.00
Femoral Heads	2.40–3.37	3.16–4.62	Femoral Heads	0.26	−0.57
Prostate	1.19–1.79	1.33–1.89	Prostate	−0.05	−0.11
Rectum	1.51–1.99	1.70–2.22	Rectum	−0.26	−0.13
Skin	1.80–1.96	2.26–2.92	Skin	0.11	−0.25
Bones	2.93–3.96	4.61–5.72	Bones	0.47	−0.77

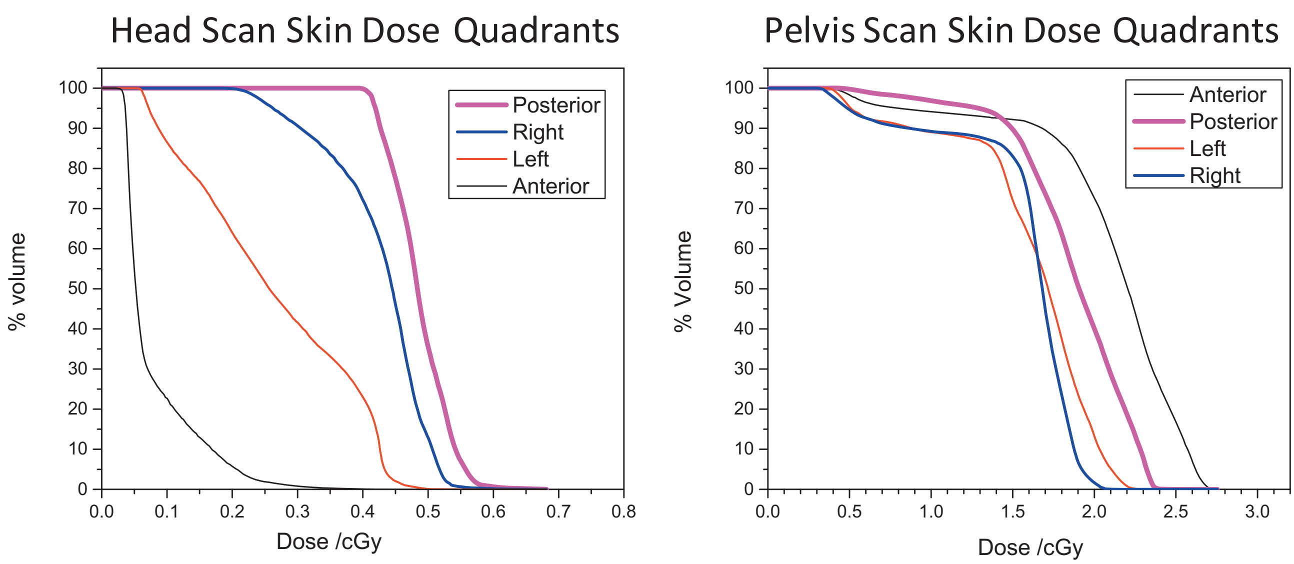

Fig. 3. Analyzing the dose–volume histogram of the four skin quadrants separately for a patient with Standard Head Scan demonstrates the positional dependence due to the 200° head scan geometry. Conversely, far less positional dependence is seen when the same analysis is applied to a Pelvic Scan due to the 370° scan geometry.

organs across all patients fell between about 1.2–2.2 cGy for D50 and 1.3–2.7 cGy for D10.

The sensitivity of organ doses on isocenter position was explored by simulating the same patient with three different isocenters: centered in the prostate and 8 cm anterior and posterior of that isocenter (16 cm total range). The results showed the greatest dose difference in the most extreme organ, the skin, where the 8 cm anterior shift resulted in a D50 of 1.3 cGy more dose to the anterior skin and 0.5 cGy less dose to the posterior skin, while the 8 cm posterior shift resulted in a D50 of 0.9 cGy less dose to the anterior skin and 1.0 cGy more dose to the posterior skin. For the most central internal organ, the prostate, the shifts resulted in D50 dose of 0.3 cGy lower with the posterior shift and 0.1 cGy lower with the anterior shift for this patient. The higher dose to the skin is due to the large decrease in SSD over some portion of the 360° scan rotation. Changes in organ dose due to these shifts are shown in Table 3b.

Skin dose dependence on scan geometry: 200° scan in Head protocol vs. 370° in Thorax and Pelvic protocols

Fig. 3 illustrates the dose to four quadrants of skin separately for a patient with Standard Head Scan, demonstrating the positional dependence due to the 200° head scan geometry and for a patient with Pelvic scan, demonstrating far less positional dependence due to the 370° scan geometry. Furthermore, organs that are relatively small or centrally located, like the prostate, tend to have more uniform dose with less inter-patient variability.

Mean and D50 organ doses

Mean organ doses are not given in the tables, but as seen in Figs. 1 and 2, were generally found to be very similar to D50 doses in most cases. The average difference between mean dose and D50 across all patients and all organs were found to be 2.6 ± 2.5% (8% max) for the Pelvic Scan, 2.1 ± 2.3% (12% max) for the Low-dose Thorax Scan, 7.7 ± 8.5% (23% max) for the Standard Head Scan for Head-and-Neck isocenters, and 6.9 ± 11.9% (58% max) for Standard Head Scan for brain isocenters. The larger variation observed in the Standard Head Scans is due to the isocenter dependence caused by the asymmetric 200° posterior scan angle. The posterior skin, for example, would receive far more dose than the rest of the skin, such that the difference between the mean skin dose (which includes the high posterior skin dose) and the D50 dose (which largely excludes the high posterior skin dose) can be over 20%.

Discussion

The following observations can be drawn from the results of this study:

1. There are clinically insignificant differences in organ dose between patients for the same scan protocol. For a given scan protocol, the soft tissue organ dose ranges demonstrate that both dose magnitude and inter-patient variation are small. For example, when each soft tissue organ is examined individually, the difference between the maximum and minimum D50 soft tissue organ dose is 0.15 cGy or lower for the Standard Head

Scan (see Fig. 1, Table 1), 0.3 cGy or lower for the Low-dose Thorax Scan (see Fig. 2, Table 2), and 0.80 cGy or lower for the Pelvic Scan (see Fig. 2, Table 3a).

2. Patient organ dose can be estimated by the scan protocol. Soft tissue D50 values were found to be an order of magnitude different between the Standard Head and Pelvic Scan, demonstrating the large dose dependence on the different scan parameters used for each protocol and the size of the patient. All dose estimates must be made using data generated for the correct scan protocol and OBI version, as the doses between scan protocols differ significantly. Note that if scan parameters are changed, then the dose received also changes. Changes in mA and ms for a given scan geometry would result in simple scaling of the doses found in this study. For kVp changes, however, these data are no longer applicable. Due to the fact that the OBI system is optimized for image quality and calibrated for each specific scan protocol, if users wish to change these values it may require commissioning a new scan protocol. Furthermore, we hope that scan protocols from other vendors will be analyzed in the future using this approach in order to have data available to clinicians who do not use Varian equipment.
3. The dependence of organ dose on isocenter position was found to be limited. Standard Head Scan D50s for almost all organs changed less than 0.1 cGy when shifted over a range of 10 cm. In the extreme example of isocenter dependence for the eyes (0.03–0.32 cGy) due to the asymmetric scan angle, the magnitude of the dose is such that this variation is not significant clinically over a course of radiation therapy. The isocenter dependence for 360° Pelvic Scan is shown in Table 3b and is less than 0.3 cGy for most soft tissue organs. The bladder was the most affected soft tissue organ (0.66 cGy lower over the 16 cm range), but this difference is in the order of the inter-patient range (1.4–2.2 cGy for bladder). Furthermore, isocenter dependence is not as important for most pelvic treatments involving kV–CBCT because they generally have a fairly comparable isocenter across patients.

Given that imaging organ dose inter-patient variation and isocenter location dependence appear to be modest, the ranges provided in the tables provide clinicians with an idea of the maximal spread that they would likely encounter in organ dose from imaging procedures across patients. It appears adequate to use the ranges found in this study as lookup values to account for and report dose from these imaging procedures over the course of a patient's treatment. For example, a typical prostate patient receiving 40 fractions of radiotherapy would almost certainly receive between 1.5–2.2 cGy per fraction extra to the rectum per pelvic CBCT. If the patient was larger, a value closer to 1.5 cGy could be assumed, or closer to 2.2 cGy per faction if the patient was smaller. Assuming a middle value of 1.8 cGy per scan, the physician could expect an extra 144 cGy to the prostate over the course of treatment if daily Pelvic CBCT were used. If CBCT scans were performed twice per fraction (pre- and post-shift), the dose would be scaled accordingly. The radiation oncologist can therefore make more informed decisions regarding scan frequency and the resulting dose to the patient.

The advantage of this approach is that it is easy to implement. In contrast to patient-specific calculations, no commissioning is necessary and very little time is required to find reasonably accurate imaging dose estimates using this method. The main disadvantage of this approach is reduced accuracy. While the given ranges will likely be valid for most patients, extreme patient geometry or scan isocenter location could result in organ doses outside of these ranges. Furthermore, in situations where organ tolerances have been reached and more accuracy is required, a patient-specific calculation might be advisable.

Conclusions

For all soft tissue organs using Varian OBI system, the ranges in D50 were shown to be 0.1–0.4 cGy for Standard Head Scan, 0.3–0.6 cGy for Low-dose Thorax and 1.2–2.2 cGy for Pelvic kV–CBCT scans. D50 ranges for individual organs are provided and are generally even tighter. These ranges are valid for a majority of patients to estimate organ dose from these scan protocols.

This study has demonstrated that inter-patient variation and isocenter dependence of organ dose is modest, and therefore it may be clinically acceptable to use tabulated values based on scan procedure and site to estimate additional doses to the organ-at-risk at the time of patient treatment planning. Clinicians can use these dose values to make informed decisions regarding how often to image during treatment. This approach also enables clinicians to report and account for organ dose resulting from multiple repeated imaging guidance procedures.

Conflict of interest

There is no conflict of interest.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.radonc.2014.05.019.

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